BIOS 6624 Project 2

Emily Kokesh

Data Analysis Plan

For Aims 1 and 2, analyses will begin with inspection of distributions, missingness, outliers, and correlations among inflammatory markers, outcomes, and candidate covariates. Primary inflammatory predictors will be TNF-α, IL-6, MCP-1, and Eotaxin-1, each modeled separately as a continuous predictor because preliminary data showed a correlation of approximately 0.93 between IL-6 and MCP-1, and fitting highly correlated markers together could yield unstable estimates and obscure individual effects. Primary outcomes will be the five episodic memory outcomes from the grant's Table 3 and mean AD-signature cortical thickness, yielding 24 models per aim.

For Aim 1, Hypothesis 1a will be tested with separate adjusted linear regression models for each outcome marker pair, with the year-1 cognitive or cortical thickness outcome as the dependent variable and the baseline inflammatory marker, corresponding baseline outcome, age, and sex as predictors. The primary hypothesis test will be a 1-df partial F test for the inflammatory marker coefficient. Hypothesis 1b will be tested with separate adjusted linear regression models in which 1-year change in each cognitive or imaging outcome is regressed on 1-year change in the inflammatory marker, again with age and sex included in all primary models. The primary inferential test will again be a 1-df partial F test for the inflammatory-marker change coefficient. Estimated regression coefficients, standard errors, 95% confidence intervals, and p-values will be reported.

For Aim 2, the primary analysis will retain amyloid burden as a continuous baseline predictor using global amyloid PET SUVR. For each outcome-marker pair, an adjusted linear regression model will include baseline amyloid SUVR, baseline inflammatory marker level, their interaction, the corresponding baseline outcome, age, and sex. The primary hypothesis test will be a 1-df partial F test for the amyloid-by-inflammation interaction term. Because both predictors are continuous, interpretation will focus on model based contrasts and predicted outcomes across low and high combinations of amyloid and inflammation rather than on the interaction coefficient alone. If the continuous SUVR and outcome relationship is not approximately linear, a secondary analysis will classify amyloid as positive versus negative using an established SUVR threshold of 1.01 and refit the interaction model.

Across both aims, additional candidate covariates will include education, APOE carrier status, cardiovascular markers, metabolic factors, BMI, hypercholesterolemia history, immune-related health history variables, Clinical Dementia Rating, Geriatric Depression Scale, NSAID use, and routinely collected cognitive measures. Continuous covariates will generally remain continuous, and clinical history variables will be coded as binary or categorical as appropriate. Covariate selection will prioritize confounders over precision variables. Highly collinear variables will not be entered together without justification, and APOE will be treated as a high priority covariate. Overall model adequacy will first be

assessed with an omnibus F test against a mode containing only an intercept, followed by prespecified partial F tests for the coefficient or interaction of interest. Model parsimony will also be assessed with AIC. Analyses of the remaining 6 cytokine/chemokine markers will be treated as exploratory. The final analytic sample for each aim will be restricted to participants with the required baseline predictor data, corresponding baseline and year-1 outcome data, and non-missing covariates for the final model. We will summarize exclusions at each step and compare included versus excluded participants on key baseline characteristics to assess potential selection bias. All analyses will be conducted in R version 4.5.3.

Power Analysis and Sample Size Justification

The planned final analytic sample is 175 participants after follow-up. Because the final analyses are adjusted linear regressions but only limited pilot data were available, power calculations were based on simplified correlation based approximations from the preliminary data.

Table 1: Aim 1 power thresholds at N = 175

Smallest representative effect size achieving at least 80% power

Type I error control	α	Minimum pilot |ρ| reaching 80% power	Power at N = 175	Required N for 80% power
Bonferroni across 48 Aim 1 tests (alpha = 0.05 / 48)	0.001	0.320	0.855	158
More strict alpha = 0.01	0.010	0.260	0.817	169
Unadjusted (alpha = 0.05)	0.050	0.260	0.936	114

Thresholds were derived from corr.1samp() results using the representative pilot effect-size range. The table complements the Aim 1 figure by summarizing the minimum detectable correlation under each type I error strategy at the planned sample size of 175.

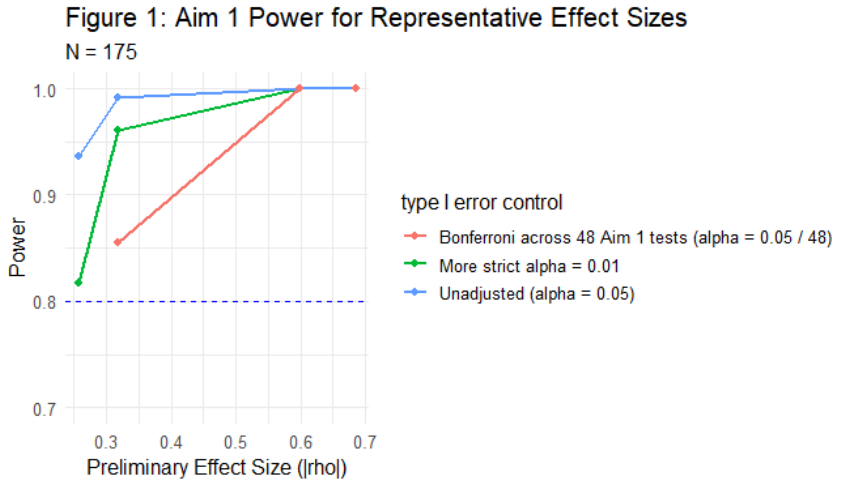

Figure 1. Power for representative Aim 1 pilot effect sizes at N = 175 under three type I error strategies.

For Aim 1, power was evaluated with corr.1samp() from the powertools package because the pilot data provided representative single predictor-outcome correlations and the main question was whether associations of plausible magnitude could be detected at N=175. We specified N=175, rho0=0, two-sided tests, alternative correlations spanning the observed pilot range, and alpha values of 0.05, 0.01, and a conservative Bonferroni correction across 48 Aim 1 tests. As shown in Table 1, Supplemental Table 1, and Figure 1, N=175 provides adequate power for moderate and larger representative Aim 1 associations, although the smallest effects become difficult to detect under conservative multiplicity control.

Table 2: Aim 2 alpha sensitivity under a representative scenario

Type I error control	α	ρ_1 amyloid+	ρ_2 amyloid-	Difference in correlations	Power at N = 175	Required total N for 80% power
unadjusted (alpha = 0.05)	0.050	−0.450	−0.250	0.200	0.309	629
more stringent alpha = 0.01	0.010	−0.450	−0.250	0.200	0.132	932
bonferroni across aim 2 primary tests (24 tests)	0.002	−0.450	−0.250	0.200	0.053	1224

Table 2 shows a representative middle-ground Aim 2 scenario under the simplified corr.2samp() approximation. The 60:40 split assumes 60% amyloid-positive and 40% amyloid-negative participants at the planned total sample size. The amyloid-negative correlation (rho2 = -0.25) was chosen as a moderate baseline association, and the difference in correlations of 0.20 implies a stronger negative association in the amyloid-positive group (rho1 = -0.45). Results are shown across alternative type I error control strategies. The full pattern across other splits, baseline correlations, and differences is shown in Figure 2 and Supplemental Table 2.

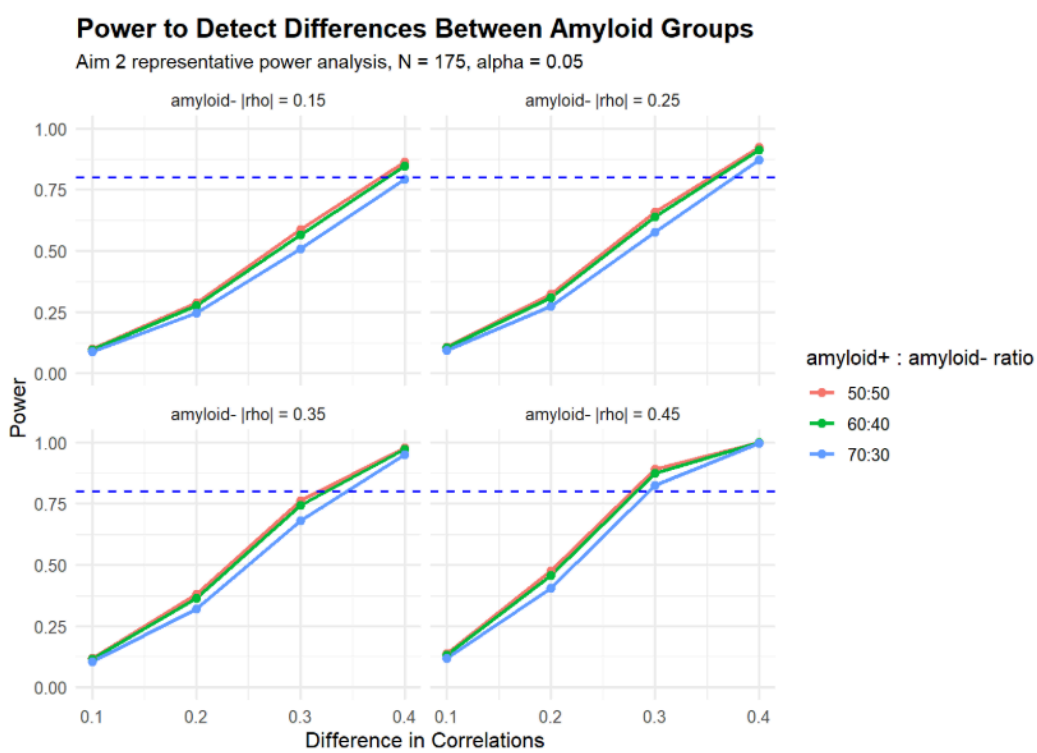

Figure 2. Aim 2 power under the simplified corr.2samp() approximation to the planned amyloid-by-inflammation interaction analysis. Because the final Aim 2 model tests whether the inflammation-outcome association differs by amyloid burden, power was approximated by comparing predictor-outcome correlations between amyloid-positive and amyloid-negative groups. Panels vary the assumed absolute amyloid-negative correlation, lines vary the amyloid+ : amyloid- split, and the x-axis shows the between-group difference in correlations. The dashed horizontal line marks 80% power.

For Aim 2, power was evaluated with corr.2samp() as a simplified approximation to the planned interaction analysis because the final model tests whether the inflammation and outcome association differs by amyloid burden. We therefore approximated the interaction by comparing predictor and outcome correlations between amyloid positive and amyloid negative groups under plausible amyloid splits. Calculations used total N=175, two-sided tests, amyloid+ to amyloid- splits of 50:50, 60:40, and 70:30, baseline amyloid negative correlations from 0.15 to 0.45 in absolute value, differences in correlations from 0.10 to 0.40, and alpha values of 0.05, 0.01, and Bonferroni correction across 24 Aim 2 tests. As summarized in Table 2 and Figure 2, Aim 2 appears underpowered for moderate interaction effects under this conservative approximation, with adequate power reached mainly under stronger scenarios with larger between group differences and stronger baseline amyloid negative correlations.

Overall, these calculations support the adequacy of N=175 for Aim 1 primary analyses but suggest more limited power for Aim 2 interaction effects. In addition to the unadjusted alpha=0.05 results, we considered alpha=0.01 as a stricter alternative attempt at type I error control without being as conservative as Bonferroni. Bonferroni-adjusted results are reported as conservative benchmarks, but because many outcomes and inflammatory markers are correlated, a multiple testing approach that accounts for dependence among tests, such as false discovery rate control or an effective number of tests approach, would likely be more appropriate for the final analysis than assuming full independence.

Supplemental Tables/Figures:

Supplemental Table 1: Aim 1 combined correlation-based power calculations

Based on representative pilot effect sizes across preliminary outcomes

Representative effect size	Type I error control	α	Pilot |ρ|	Power at N = 175	Required N for 80% power
0.26	Unadjusted (alpha = 0.05)	0.050	0.260	0.936	114
0.32	Unadjusted (alpha = 0.05)	0.050	0.320	0.991	74
0.60	Unadjusted (alpha = 0.05)	0.050	0.600	1.000	18
0.69	Unadjusted (alpha = 0.05)	0.050	0.690	1.000	13
0.26	More strict alpha = 0.01	0.010	0.260	0.817	169
0.32	More strict alpha = 0.01	0.010	0.320	0.961	109
0.60	More strict alpha = 0.01	0.010	0.600	1.000	26
0.69	More strict alpha = 0.01	0.010	0.690	1.000	19
0.26	Bonferroni across 48 Aim 1 tests (alpha = 0.05 / 48)	0.001	0.260	0.579	245
0.32	Bonferroni across 48 Aim 1 tests (alpha = 0.05 / 48)	0.001	0.320	0.855	158
0.60	Bonferroni across 48 Aim 1 tests (alpha = 0.05 / 48)	0.001	0.600	1.000	38
0.69	Bonferroni across 48 Aim 1 tests (alpha = 0.05 / 48)	0.001	0.690	1.000	26

Power calculations were conducted using corr.1samp() from the powertools package. Representative effect sizes were taken from the observed pilot correlation range across preliminary memory and cortical thickness outcomes.

Reproducible Research:

- **GitHub repository:** https://github.com/emkokesh/BIOS6624
- **Raw preliminary dataset:** C:/Users/kokeshem/Spring 2026/Advanced Data Analysis/BIOS6624/Project2/DataRaw/PrelimData.csv
- **Analysis script:** C:/Users/kokeshem/Spring 2026/Advanced Data Analysis/BIOS6624/Project2/Code/Power_Analysis.Rmd
- **Rendered analysis file:** C:/Users/kokeshem/Spring 2026/Advanced Data Analysis/BIOS6624/Project2/Code/Power_Analysis.html
- **Final report**: C:/Users/kokeshem/Spring 2026/Advanced Data Analysis/BIOS6624/Project2/Reports/Bios_6624_Project2.pdf